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Communication 494LI

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An Algorithmic Audit: TikTok's Algorithmic Control Rooted in Personalization

This algorithmic audit discovers how TikTok extracts user information to personalize content and shape usability. TikTok uses a predictive recommendation system that takes users behavioral data and uses it for personalization, allowing the algorithm to control engagement and opacity. Algorithmic personalization can lead to bias, and we use TikTok just as it uses us. There is an interplay between how we are entertained with the app and how we feed our personalities into it as it creates more incentive for us to stay connected for TikTok's gain of power. The power of surveillance is at the forefront of my audit. Automated surveillance is dangerous in any capacity and can be used to influence users any way the app sees fit. TikTok's predictive algorithmic system allows for a powerful method of personalization while increasing their ability to control users' actions.

In conducting this audit, my goals for the study were to discover how two different TikTok accounts interpret different profiles of users and personalize their content. Throughout my research I also wanted to discover how the design of the app prompts users to do certain things. I became more interested in how the user experience power of TikTok intertwines with the personalized content to keep users as engaged as possible. This led me to investigate how TikTok's interface and data collection work in a sociotechnical system to increase user engagement. I primarily wanted to analyze how these factors introduce bias and shape what users

see and how they interact with the app. By conducting my study using two different TikTok accounts, each with different preferences and niche interests, pop-culture and travel, I was able to see what content overlapped and how each accounts algorithmic preferences responded to searches. In addition to monitoring content on TikTok's "For You page" (FYP), I took note of each videos suggested searches and become interested in how content is being pushed through different sections. Guiding my audit were a few questions; why does TikTok choose to use some of our information over others? Do the two accounts have some similar content or similarities in targeting and what is the overlap? What does the content and prompting searches inform about the motivations of TikTok? The power of TikTok lies within its predictive control and the algorithm's model of using user data in automated monitoring to produce personalized content.

For my audit, I set up two different profiles on TikTok. One was my personal account, I follow celebrities, friends and pop-culture. The other account is a niche account for traveling and nature for which I followed other travel creators and posted travel content. TikTok is a popular social media platform that consists of attention grabbing, short-form videos posted by creators. The app focuses on creator-user interaction and operates on an observational algorithm to customize users' experiences and content through their preferences. TikTok's specific algorithm functions as a predictive model and makes assumptions about the users while it makes predictions about future behavior and preferences. During my audit I gathered data and information through daily usage and observation on both accounts, observations of others and survey data. To collect my own data and observations, I set up a schedule – two sessions a day, one in the morning and one at night, each consisting of fifteen minutes. To make sure I was receiving accurate information, I made sure to watch ten videos in their entirety on each account, like five videos and conduct one search. I conducted this observation for fourteen days straight,

recording my observations and tracking the changes that occurred as well as the overlapping content on each profile. I took screenshots of searches, results of searches and content that filtered through each profile's FYP. I conducted an interview-style survey where I got results from thirteen participants with nine questions regarding their activity on TikTok. This methodology allowed for a feasible study to be conducted within the timeframe I had to work on it. I made sure to collect enough data to be able to see and understand how TikTok operates as a predictive model and works to optimize user engagement as well as their own motivations. By collecting this information, I worked through my thesis and curated a better understanding of what I wanted to learn from this study.

Throughout my audit, I collected data that was both surprising and some of what I had predicted. Both accounts offered insight as to how our preferences and demographics on TikTok influence the results of the preferential algorithmic output. There is an evident correlation between personalization and algorithmic control. When I searched for content on each of the profiles I created there were diverse results in output. I did many trials in which I searched the same phrase or word on each of the two accounts, then noted which types of videos came up in the results. I also began to note the suggested searches for each account which further prompts users to interact with this function. By comparing a controlled search on each profile, I could come to conclusions as to how each profile was being seen by the algorithm. To stay within the realm of neutrality, I focused my searches on trends and non-ideological phrases. The searches I conducted included "Work", "My Type", "Day in my life (DIML)", and "Travel". When I looked up "work" on both profiles, very different content was given to me. Figure 1 shows the results of "work" received from the search on my main, pop-culture account. This search resulted in corporate and marketing work. All of the videos under this search had to do with working in an

office and are in the subject of work I am personally most interested in or am getting a degree in. Figure 3 is a result on my main of a “workday in my life” which is a corporate office job. When I did this same search on my travel account, very different results showed up. The suggested videos under this section can be seen in figure 2. The videos are jobs in a more casual setting. All of the people in the videos are wearing sweatshirts and most of them are in the service industry. Figure 4 is a result of content related to “workday in my life”, a big switch from the corporate workday from my other account. As I know TikTok does profile users and tries to keep users as engaged as possible through algorithmic preferences, it makes sense that the algorithmic output of these searches seems to generalize the professionalism of the users. The pattern of these findings reinforces TikTok’s algorithmic categorization and how this translates into making assumptions about user’s lifestyles based on their behavioral data. There is also an obvious assumption about users’ occupation and general interest regarding employment and geolocation. Making these assumptions only influences and reinforces user behavior as it influences their inherent self-biases. While these searches are surface level, they reveal that external biases are made which influence internal biases. In another search, I thought it would be interesting to search for “My type” on each account. This is part of a trend in which you are supposed to get results for someone who is your type of love interest. My initial thoughts were that the results would not vary too much because the two profiles I am working with are not all that different. But the differences are telling as to how predictive and thoughtful the model is. Figures 5 and 7 are the results of the search “My type” on my main, pop-culture account. Figures 6 and 8 are the results from my travel account. The output on both accounts is similar at first glance, even having the same video within the top four results. The output for my personal account are of traditionally handsome, educated presenting men. All of the videos are of men that are well

dressed and assumed to be well off. The output for my travel account are generally of men with a different set of skills. While some of the videos overlap, the difference is that the travel profile receives a distinctly lower class of interest. On further inspection, the travel account has a consistent theme of “blue collar”, conservative nature among the men while my personal account has an emphasis on “white collar”, liberal looking men. The algorithms seem to have made the assumption that my personal, pop-culture account would respond better to creative liberalism and corporate stereotypes while the more niche travel account would respond to more conservative, blue collar content. Outside of testing the searches for each TikTok account, I also observed changes in content on my FYP. TikTok’s algorithms appear to be highly adaptive to content that users seem to enjoy. Throughout my observation sessions twice a day, I noticed several times when I watched the entire video from a creator I have never seen before, another one of their videos shows up two videos later. This tells me that TikTok’s predictive algorithm works extremely fast to adapt in real time with each one of its users according to their interests. The speed and accuracy reveal how TikTok’s recommendation system maximized engagement with the necessary speed. This explains why their model works so well in keeping users engaged for such a substantial amount of time.

In addition to conducting my own observations, I surveyed others to further see their experiences. To collect a larger number of answers in a low-pressure way, I used an online survey I created using Google Forms. I asked nine questions pertaining to TikTok users’ tendencies when using the app, their interests and a self-report of their overall thoughts the app. Many of the questions I asked had to do with opacity and how users understand how they are being manipulated or not. The results of this survey found that 100% of the participants stayed on TikTok for longer because they receive content that is more relevant to them (Figure 10).

However, when asked how engaged users were on a scale from 1-10, the responses varied (Figure 9). This indicates that users feel much more engaged and are likely to stay connected to TikTok when they are targeted with content based on their preferences. Users also don't feel like they are as engaged as they actually are, indicating intentional opacity of the intentions of TikTok's targeting algorithm.

As collected in my survey data, TikTok is a highly influential platform in today's society. The content we see on the app contributes to how much time spent on it and reflects our interests. However, even though people seem to understand this causal relationship it is intentionally difficult for them to recognize all the different factors that contribute to their engagement on TikTok. Throughout my observations and surveying I reconfirmed my claims that TikTok uses surveillance to personalize their content. It is an extractive process in which we as users are having our data taken. The platform itself is designed to prompt users to search and be as entertained as possible while working within its own agenda. These findings suggest that surveillance is a source of great power. What users fail to understand, intentionally, is that they are not only consumers of TikTok, but they are their entire business model and source of data. What has been artfully framed as personalization, even by me, is automated surveillance. People have become used to being surveilled, which makes TikTok and its wide reach so influential and gives it the potential to be dangerous. In automated surveillance, information and data is stored indefinitely and forever to be used at any given time. When conducting my searches and making observations, I learned that TikTok actively changes its mind based on our minds. Everything works together to take user preferences and curate the most digestive flow of content. Even slight differences between search results or suggestions prove that the algorithms understanding of the individual is vast. The more users give their information, the more accurate it becomes in return.

My research was more surface level in content format but reveals a heavier truth; predictive algorithmic structures have the power to manipulate users because they know them so well. Platforms like TikTok are described as being neutral and not bias, but everything they work with points to the opposite. Ideologies of users who prefer certain topics can easily be exploited with how content it distributed on TikTok. Content on any social media has the potential to shift one way or another. As traditional media becomes more obsolete, social media picks up the slack of the public's attention when it comes to politics and news. My research has proven just how accurately TikTok's predictive algorithm is able to use preferential data for entertainment, but what does this mean in a wider lens? TikTok is able to accurately interpret each person's profile to form an accurate predictive output for entertainment purposes. There is also a possibility for this powerful system to be used in a more cynical way to push ideological or political ideas to users the algorithm thinks could be receptive.

These ideas are some that I would have liked to have studied more, especially after writing this report and having time to synthesize my thoughts. There were some aspects of my research and the process of my audit that I could have improved on. After developing a deeper theoretical understanding of the power of social medias algorithmic systems in larger work issues, I would be fascinated to study algorithms on different platforms. In the future, I would find it interesting to investigate the difference in influence of Instagram as it compares to TikTok. It would be interesting to similarly do an algorithmic audit during a major political campaign or event. I would have found it extremely interesting to frame this audit around political bias and profiling during the 2024 U.S. Presidential Election. However, I could make assumptions as to what these biases were and how they influenced the results of the election. Another major challenge of my study was time. I changed my audits topic after the first draft was completed and

lacked the time to organize my new one without as much feedback. I feel as though this audit was much more informative, however. I was still able to get two weeks of observation and data collection which gave me enough information. The more I conducted this study and learned from what I saw, I found myself wanting more time to change some aspects or try new things and talk to more people. I would similarly have liked to have talked with more people in my survey data. I had people I know well, which inevitably have similar habits as me, respond to my survey. I would have liked to have more people who I do not know as well take my survey to get a more diverse response. The survey would have also been more useful if I had included more questions or had the time to do multiple surveys to each group. There could have have been deeper questions included on the survey I conducted that would have given me more insight into people's thoughts. Similarly, I would have liked to have spoken to more people in person to have had more of a conversation with them. This would also have allowed me to observe people's behaviors while using the platform, as opposed to simply recounting it. This all would have allowed me to deepen my research and strengthen my argument.

When conducting this study and thinking about the General Education courses that I have taken over the years, I immediately thought of my Psychology 100, Introduction to Psychology class. This class was entirely influential for me and inspired me to pursue a minor in Psychology. I find learning about people and their motivations fascinating. Learning about this helped me interpret why algorithms are developed certain ways to satisfy people's needs. The psychology of humans and how our minds work is very informative as to how an algorithmic system relies on behavioral data. In thinking about TikTok, their algorithm is predictive in order to keep us as engaged and satisfied as possible. The interplay between algorithmic power and success with human psychology is undeniable. This psychology course helped me interpret the sociotechnical

aspect of TikTok's algorithmic control. Understanding what users want, how to get it from them and how to use that information for their gain is crucial to understanding TikTok's system of operation.

TikTok's predictive algorithm takes user information to create a personalized, predictive output. Preferences are disguised as personalization just as algorithmic control is disguised as personalization. Users are at the mercy of their own identities and information that has been extracted through their data by TikTok. This not only leads to bias through general classifications, but to a total increase of power for TikTok. Through automated surveillance and preferencing, TikTok has intentionally disguised their domination of power. The unknown of it all lies within the future. User data is stored forever, along with their curated profiles and shaped identities. My study has revealed that we as users are already gripped by the power of TikTok's entertainment. This could be used in a means to a much different end. Personalization is indefinitely a form of control to be used in anyway TikTok sees fit. Exploitation and abuse of this power are inherently possible and intentionally hidden under a guise of entertainment.

References

Figure 1

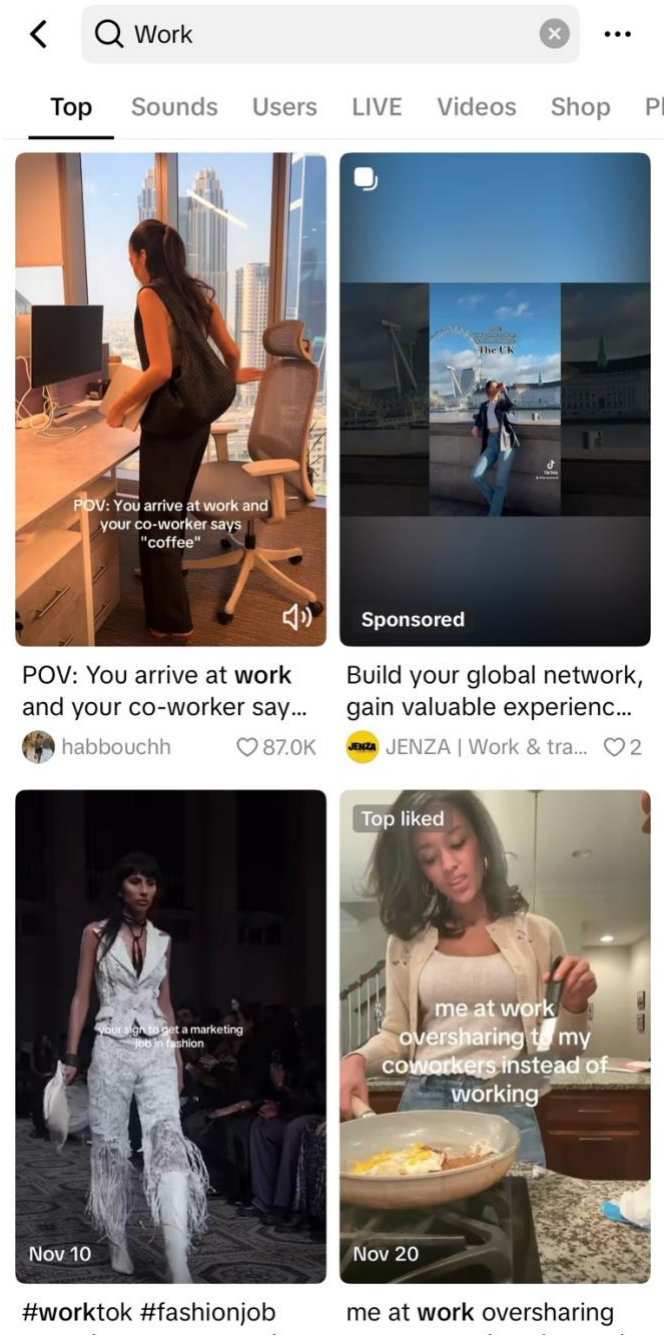

Figure 2

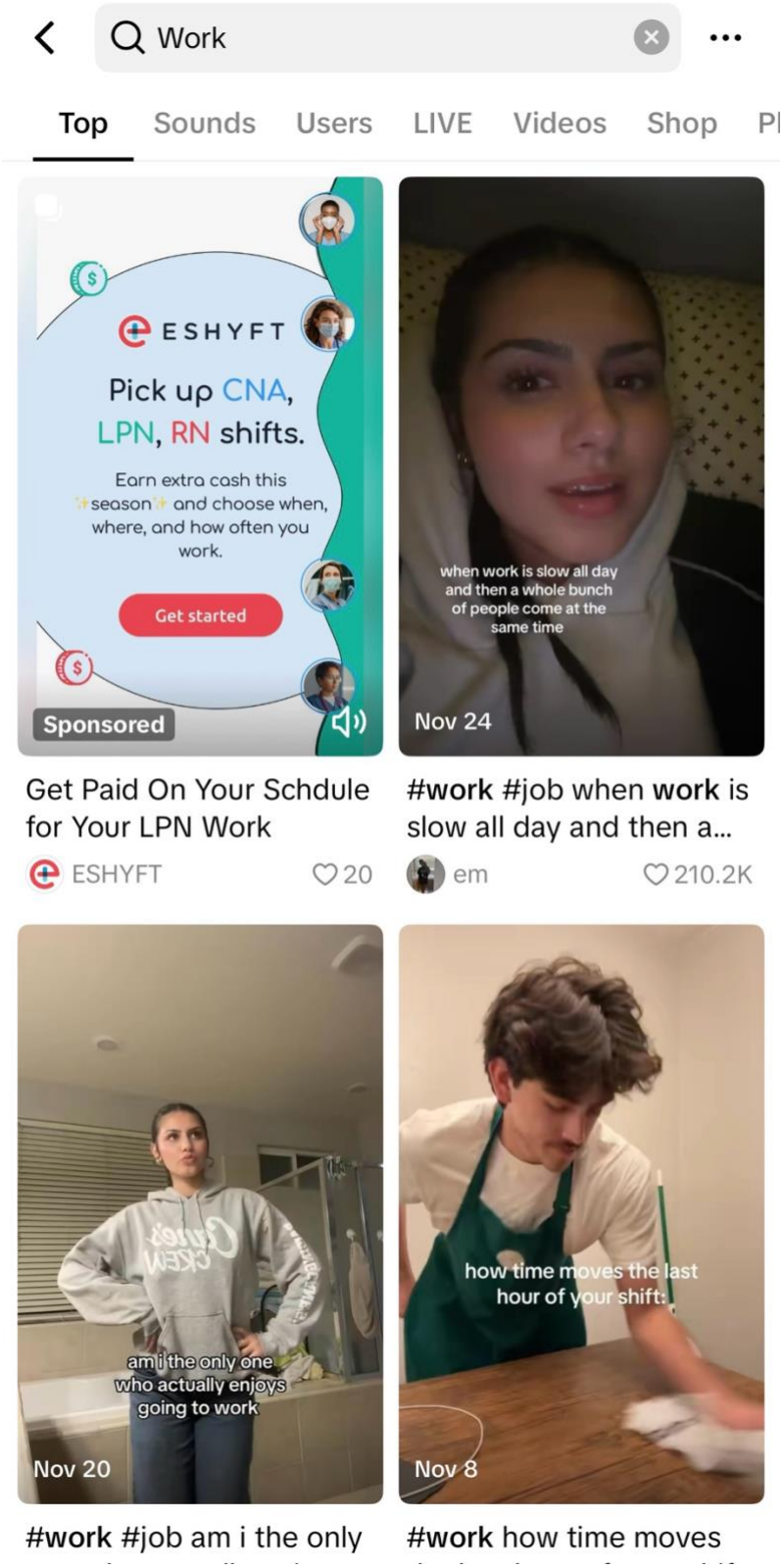
Work
Top
Sounds
Users
LIVE
Videos
Shop
ESHYFT
Pick up CNA, LPN, RN shifts.
Earn extra cash this season and choose when, where, and how often you work.
Get started
Sponsored
Get Paid On Your Schdule for Your LPN Work
ESHYFT
20
when work is slow all day and then a whole bunch of people come at the same time
Nov 24
#work #job when work is slow all day and then a...
em
210.2K
am i the only one who actually enjoys going to work
Nov 20
#work #job am i the only
how time moves the last hour of your shift:
Nov 8
#work how time moves

Figure 3

Work Day In My Life
Search
a corporate girlie
Day in the Life
2,257
22
142
25
Southern California
Zoe · 11-24
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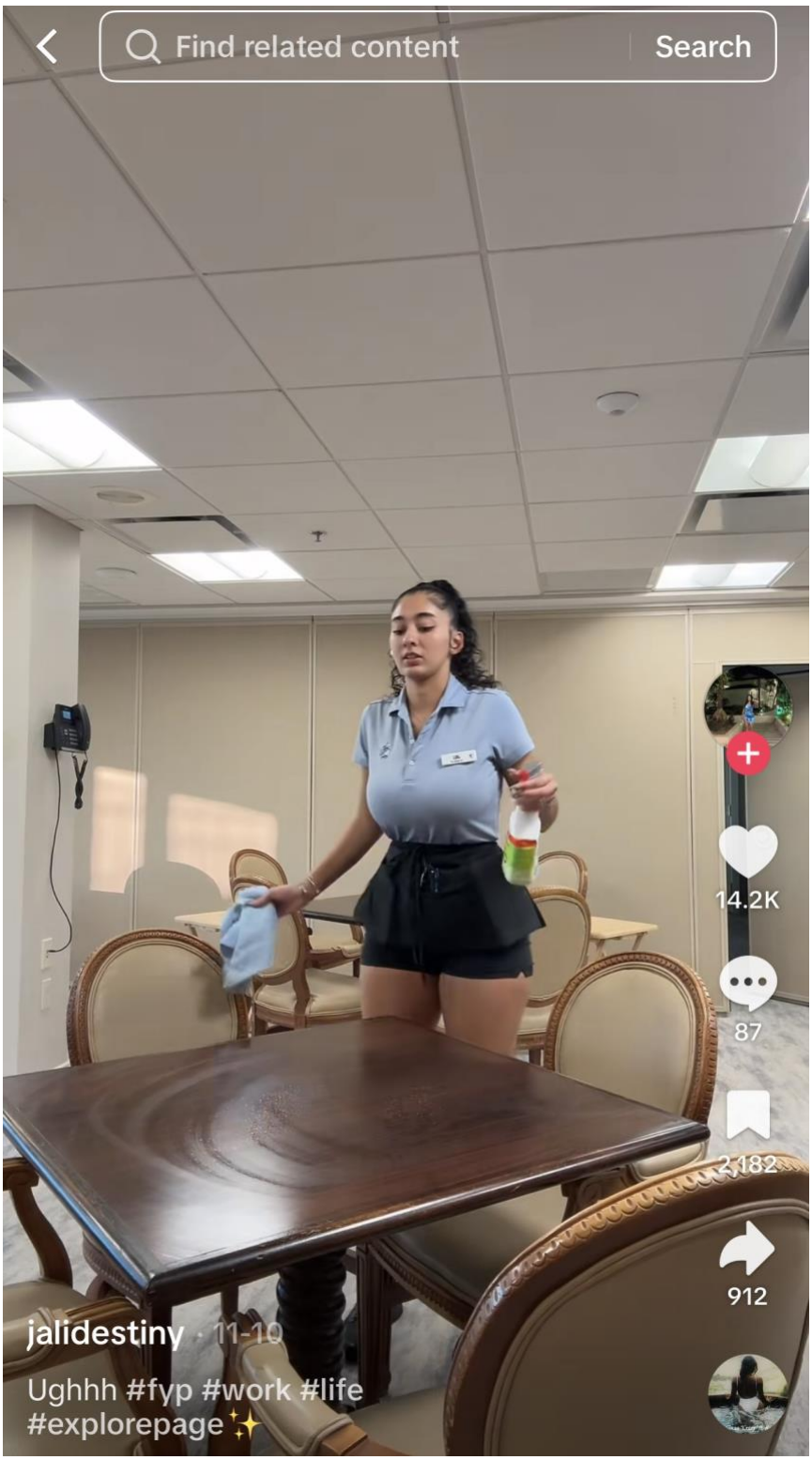
Playlist · Day in the Life

Figure 4


Find related content
Search
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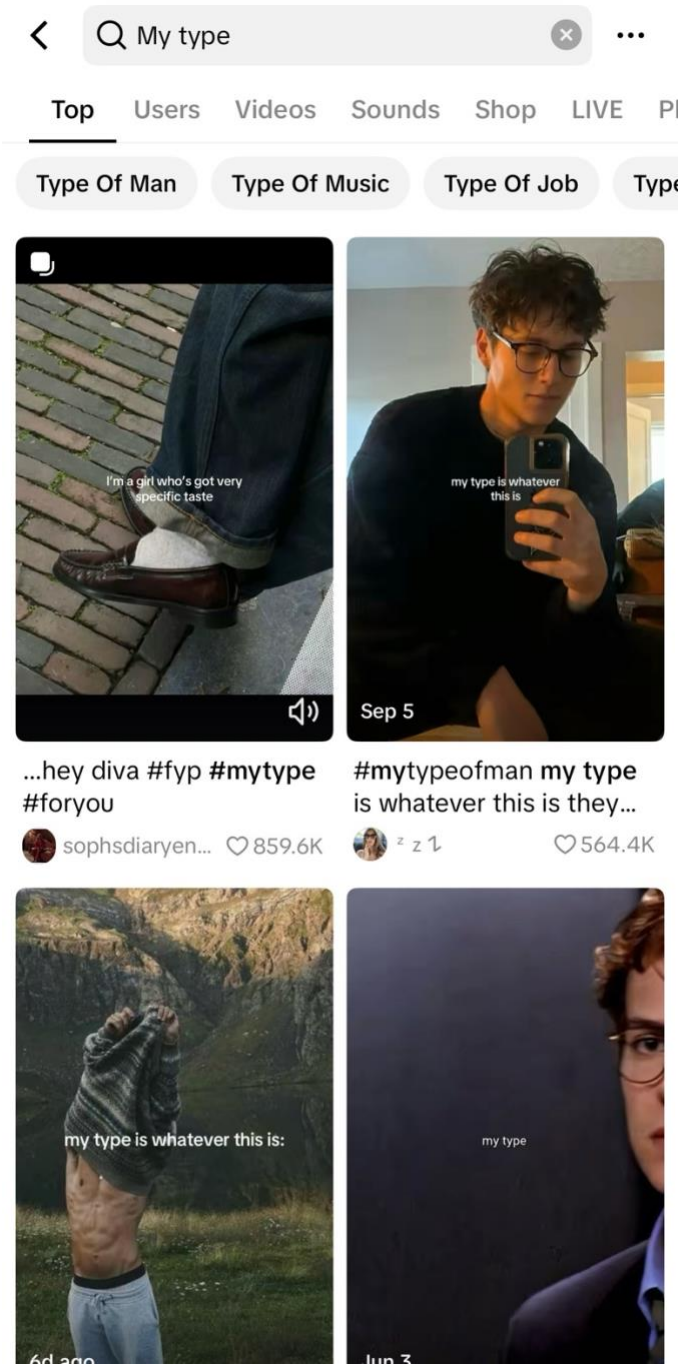
Ughhh #fyp #work #life
#explorepage✨

Figure 5


My type
Top
Users
Videos
Sounds
Shop
LIVE
Type Of Man
Type Of Music
Type Of Job
I'm a girl who's got very specific taste
my type is whatever this is
Sep 5
...hey diva #fyp #mytype #foryou
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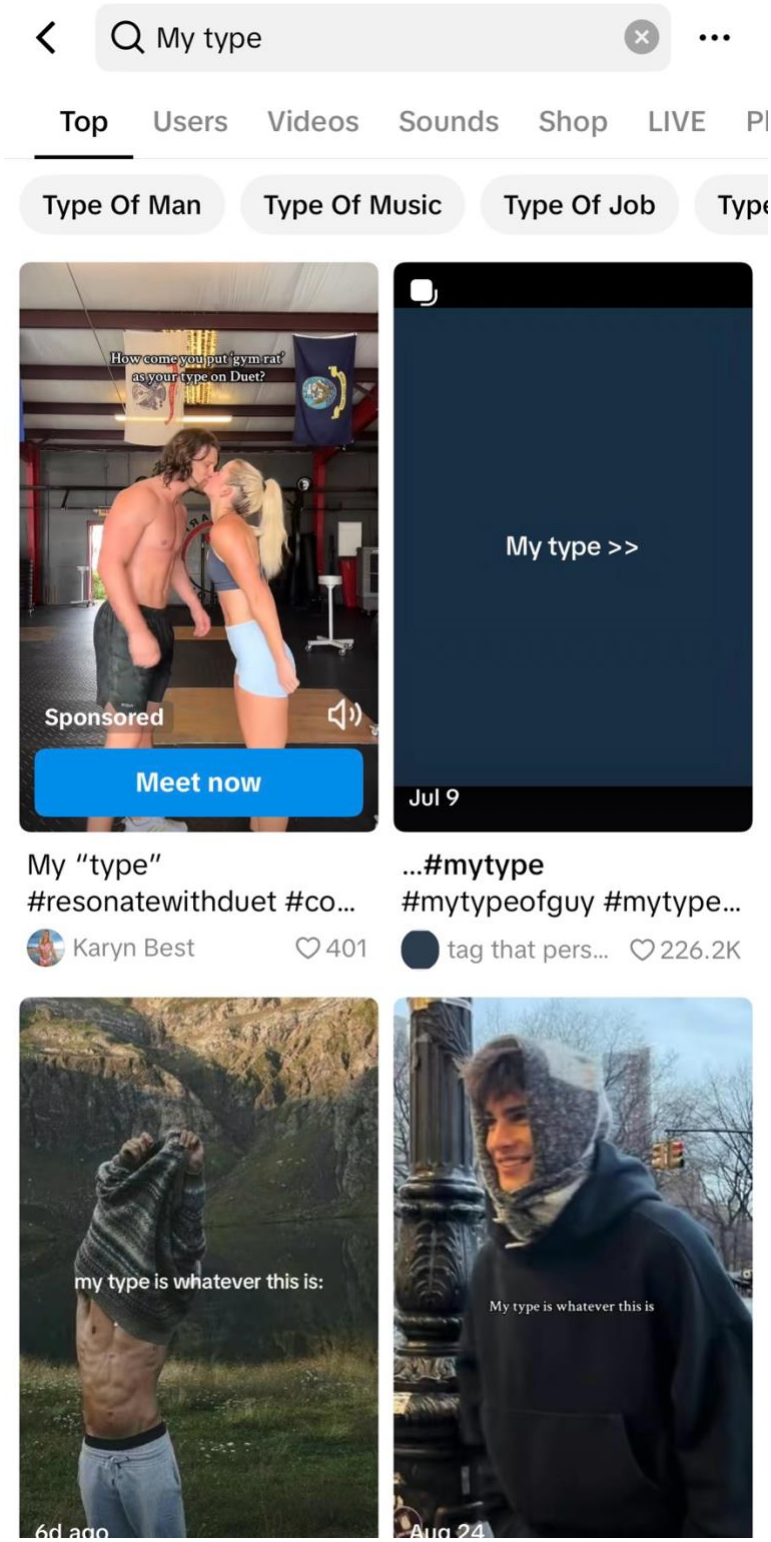
my type is whatever this is:
my type

Figure 6


My type
Top
Users
Videos
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Shop
LIVE
Type Of Man
Type Of Music
Type Of Job
How come you put 'gym rat' as your type on Duet?
Sponsored
Meet now
My type >>
Jul 9
My "type" #resonatewithduet #co...
Karyn Best
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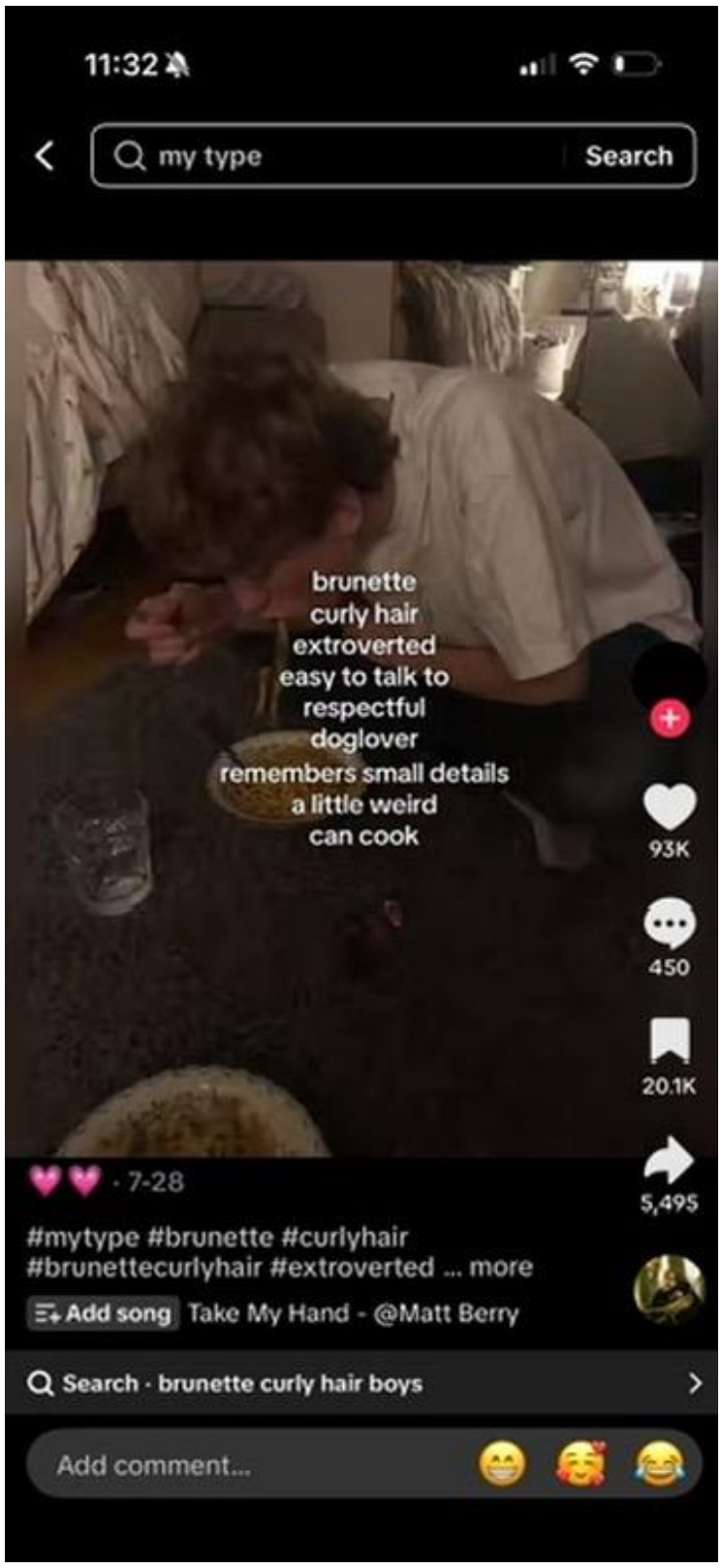
6d ago
My type is whatever this is
Aug 24

Figure 7


11:32
my type
Search
brunette
curly hair
extroverted
easy to talk to
respectful
doglover
remembers small details
a little weird
can cook
93K
450
20.1K
5,495
· 7-28
#mytype #brunette #curlyhair
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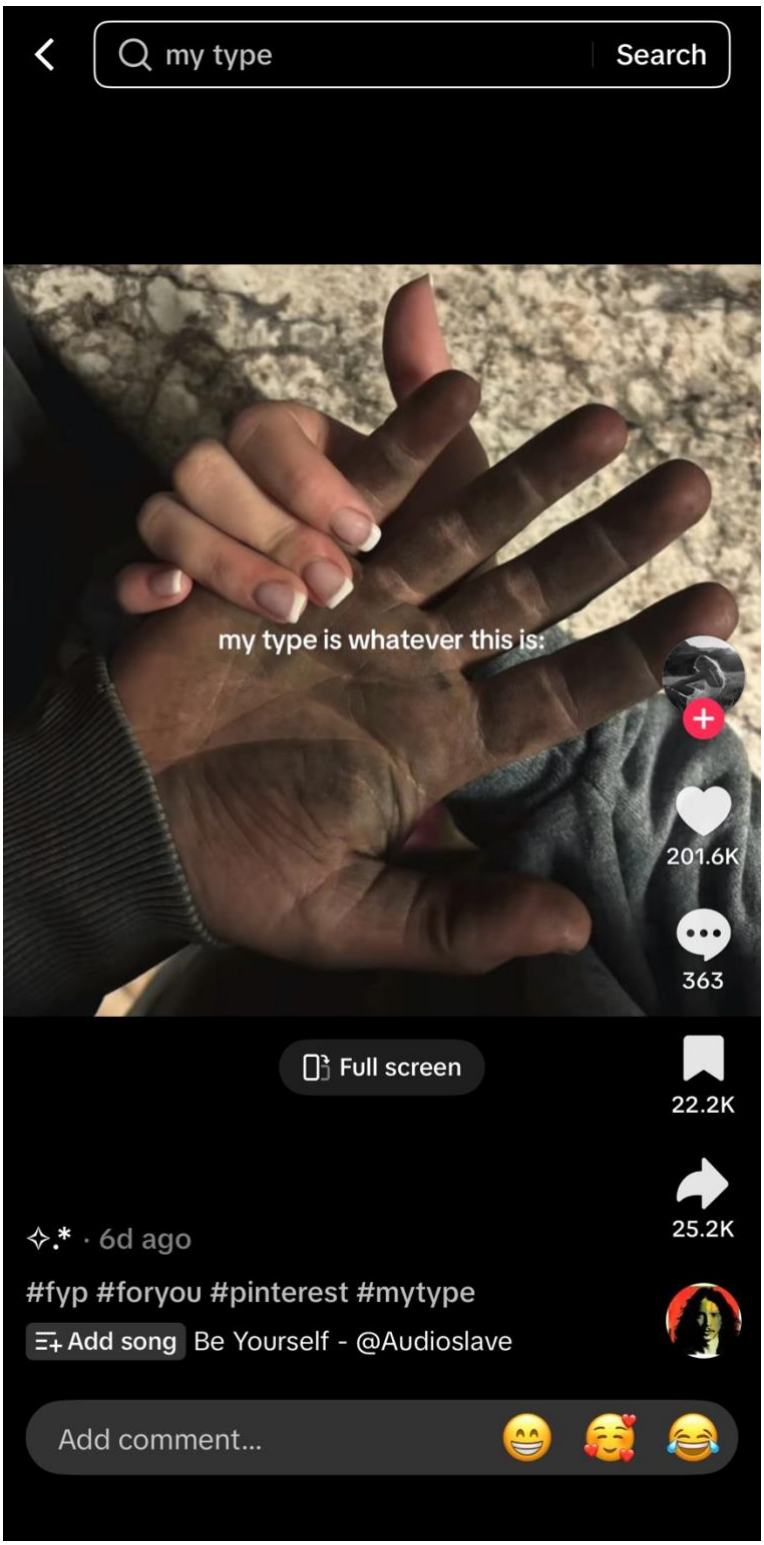
Add song
Take My Hand - @Matt Berry
Search · brunette curly hair boys
Add comment...

Figure 8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Search
my type is whatever this is:
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Full screen
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Add song
Be Yourself - @Audioslave
Add comment...

Figure 9

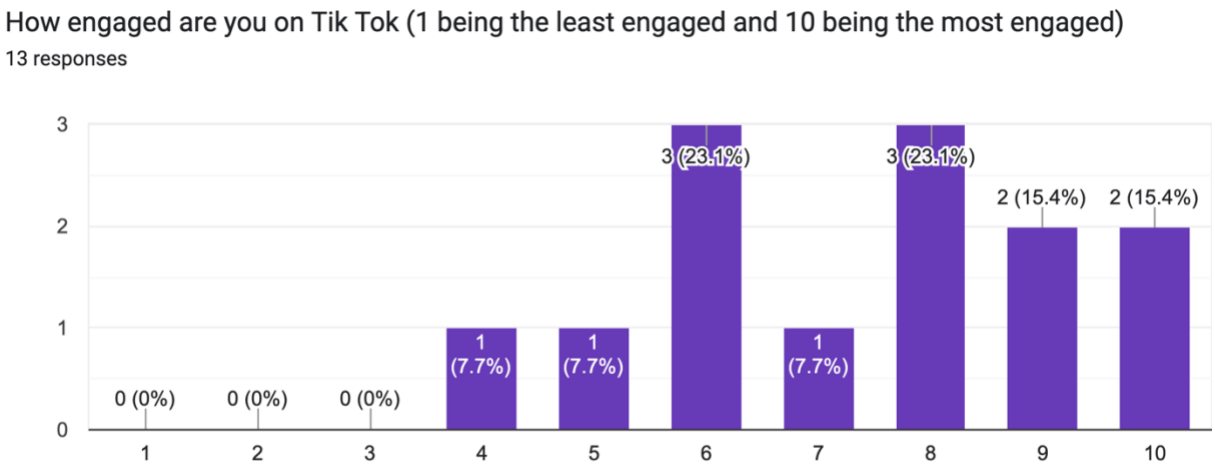
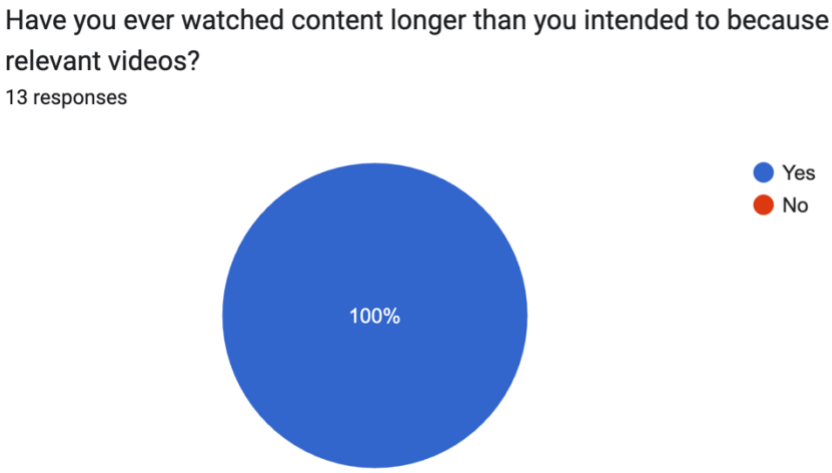


Figure 10



Survey Data:
https://docs.google.com/spreadsheets/d/1OKHko6p8fGHIL2CYRCEaJq71a6J3735oW0Vmsz_tAa8/edit?usp=sharing

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